

2023.12

$$1. \text{cond}_2(A) = \sqrt{\frac{\lambda_{\max}(A^T A)}{\lambda_{\min}(A^T A)}} = \frac{61}{\sigma_1}$$

$f(x)$ 是 $\mathbb{R}^n$ 上的范数

$$2. \begin{pmatrix} 4 & 1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -\frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} 2 & -\frac{1}{2} \\ \frac{\sqrt{3}}{2} & 2 \end{pmatrix}$$

$$3. \text{cond}_2 A = 4 \quad \|A\|_F = \sqrt{26}$$

$$4. \frac{n^3}{3}$$

$$5. a = \frac{1}{3} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \sum_{k=0}^{\infty} (a a^T)^k =$$

$$\frac{1}{2} A = a a^T = \begin{pmatrix} \frac{2}{3} \\ -\frac{1}{3} \end{pmatrix} \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \end{pmatrix} = \begin{pmatrix} \frac{4}{9} & -\frac{2}{9} \\ -\frac{2}{9} & \frac{1}{9} \end{pmatrix}$$

$$\sum_{k=0}^{\infty} A^k = (I - A)^{-1} = \left[\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} - \begin{pmatrix} \frac{4}{9} & -\frac{2}{9} \\ -\frac{2}{9} & \frac{1}{9} \end{pmatrix} \right]^{-1}$$

$$= \begin{pmatrix} \frac{5}{9} & \frac{2}{9} \\ \frac{2}{9} & \frac{8}{9} \end{pmatrix}^{-1} = \begin{pmatrix} 2 & -\frac{1}{2} \\ -\frac{1}{2} & \frac{5}{4} \end{pmatrix}$$

$$6. f(x) = 9x^2 - 8x + 5 \mid x=0$$

$$f[0, 0] = 5, \quad f[2, 3, 4, 5, 6] = 0$$

$$7. A = \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix} \quad A = QR \quad \text{求 } R$$

$$A = (a_1, a_2) \quad a_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \quad \|a_1\|_2 = 5$$

$$w = a_1 - \|a_1\|_2 e = \begin{pmatrix} 3 \\ 4 \end{pmatrix} - 5 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

$$Q = H(w) = I - \frac{2}{w^T w} w w^T w = 2I$$

$$QA = \left(I - \frac{2}{w^T w} w w^T \right) A$$

$$= A - \frac{2}{w^T w} w w^T A$$

$$= \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix} - \frac{1}{10} \begin{pmatrix} -20 & 40 \\ 40 & -80 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} = R$$

$$8. 3x^3 + x^2 + 6x + c - x^2 - 6x^2 - 5x - 4 = 2(x-2)^3 = 2x^3 - 12x^2 + 24x - 16$$

$$a=13 \quad b=29 \quad c=42$$

$$9. |\phi'\sqrt{5}| < 1 \Rightarrow \beta \text{ 取值范围 } (0, \frac{1}{\sqrt{5}})$$

$$\text{若要局部收敛到 } x^* = \sqrt{5} \Rightarrow \phi'\sqrt{5} \neq 0 \quad \phi''\sqrt{5} \neq 0 \quad \beta = \frac{1}{2\sqrt{5}}$$

$$10. \underline{(-2, -1)(0, 1)(3, 4)}$$

$$x^3 - 2x^2 - 6x + 3 = 0$$

$$f(0) = 3 > 0$$

$$f(1) = -4 < 0$$

$$f(2) < 0$$

$$f(3) < 0$$

$$f(4) > 0$$

$$f(-1) > 0$$

$$f(-2) < 0$$

$$11. \underline{6} \quad 2x^4 - 5x + 9 \big|_{x=1} = 6$$

$$12. u_{n+1} = u_n + \frac{h}{2}(u_n' + u_{n+1}') = u_n + \frac{h}{2}(-2u_n + 2t_n^2 - 2u_{n+1} + 2t_{n+1}^2)$$

$$\text{二. } \det(\lambda(D-L) - U) = \begin{vmatrix} 2\lambda & 1 & 1 \\ \lambda & \lambda & 1 \\ \lambda & \lambda & -2\lambda \end{vmatrix} = -\lambda(2\lambda+1)^2 = 0$$

$$\rho(B) < 1 \quad \text{收敛}$$

$$\begin{cases} x_1^{(k+1)} = \frac{1}{2}(b_1 + x_2^{(k)} - x_3^{(k)}) \\ x_2^{(k+1)} = b_2 - x_1^{(k)} - x_3^{(k)} \\ x_3^{(k+1)} = -\frac{1}{2}(b_3 - x_1^{(k)} - x_2^{(k)}) \end{cases}$$

$$\Xi. A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & -6 \\ 3 & 6 & 10 \end{pmatrix}$$

① LU分解

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & -6 \\ 3 & 6 & 10 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & & \\ 2 & 1 & \\ 3 & 0 & 1 \end{pmatrix}}_L \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ & 1 & -12 \\ & & 1 \end{pmatrix}}_U$$

② 反幂法 $U^{(K)} = [1, -0.5, -0.5]^T$

求 λ, U^{KH}

$$V^{KH} = A^{-1} U^{(K)}$$

$$= \begin{pmatrix} 86 & -2 & -27 \\ -38 & 1 & 12 \\ -3 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -0.5 \\ -0.5 \end{pmatrix}$$

$$= \begin{pmatrix} 100.5 \\ -44.5 \\ -3.5 \end{pmatrix}$$

$$\lambda \approx \frac{1}{\max(V^{KH})} = \frac{1}{100.5}$$

$$U^{KH} = \frac{V^{KH}}{\max(V^{KH})} = \begin{pmatrix} 1 \\ -\frac{44.5}{100.5} \\ -\frac{3.5}{100.5} \end{pmatrix}$$

$$A^{-1} = (X_1, X_2, X_3)$$

$$AX_1 = e_1 \quad AX_2 = e_2 \quad AX_3 = e_3$$

$$LY_1 = e_1 \quad UX_1 = Y_1$$

$$\begin{pmatrix} 1 & & \\ 2 & 1 & \\ 3 & 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \begin{cases} a=1 \\ b=-2 \\ c=-3 \end{cases}$$

$$Y_1 = (1, -2, -3)^T$$

$$\begin{pmatrix} 1 & 2 & 3 \\ & 1 & -12 \\ & & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix} \quad \begin{cases} a = 86 \\ b = -38 \\ c = -3 \end{cases}$$

$$x_1 = (86, -38, -3)^T$$

$$\text{同理可得 } x_2 = (-2, 1, 0)^T \quad x_3 = (-27, 12, 1)^T$$

$$\text{四. 至少=3阶} \quad c_0 = c_1 = c_2 = 0 \quad c_3 \neq 0$$

$$a_0 \quad a_1 \quad a_2 \neq 0 \quad \beta_0 = 0 \quad \beta_1 = \frac{4}{5} \quad \beta_2 = \frac{2}{5}$$

$$l_0 = a_0 + a_1 + a_2 = 0 \quad a_0 + a_1 + 1 = 0$$

$$l_1 = a_1 + 2a_2 - (\beta_0 + \beta_1 + \beta_2) = 0 \quad a_1 + 2 - \left(\frac{4}{5} + \frac{2}{5}\right) = 0$$

$$l_2 = \frac{1}{2}(a_1 + 2^2 a_2) - (\beta_1 + 2\beta_2) = 0 \quad \frac{1}{2}(a_1 + 4) - \left(\frac{4}{5} + \frac{4}{5}\right) = 0$$

$$\text{解得 } a_0 = -\frac{1}{5} \quad a_1 = -\frac{4}{5} \quad \therefore p(\lambda) = \lambda^2 - \frac{4}{5}\lambda - \frac{1}{5}$$

$$b(\lambda) = \frac{2}{5}\lambda^2 + \frac{4}{5}\lambda$$

$$(2) \quad c_3 = \frac{1}{6}(a_1 + 2^3 a_2) - \frac{1}{2}(\beta_1 + 2^2 \beta_2) = 0$$

$$c_4 = \frac{1}{24}(a_1 + 2^4 a_2) - \frac{1}{6}(\beta_1 + 2^3 \beta_2) = -\frac{1}{30}$$

$$\therefore \text{局部截断误差主项为 } -\frac{1}{30} h^{(4)} w^{(4)}(t_n)$$

$$p(\lambda) = \lambda^2 - \frac{4}{5}\lambda - \frac{1}{5} = (\lambda - 1)(\lambda + 1) = 0 \quad \lambda_1 = 1 \quad \lambda_2 = -1$$

满足根条件, $\rho = 3 > 1$ 故此格式收敛

$$(3) \quad p(\lambda) = \bar{h} p(\lambda) = \lambda^2 - \frac{4}{5}\lambda - \frac{1}{5} - \frac{2}{5}\lambda^2 \bar{h} - \frac{4}{5}\lambda \bar{h}$$

$$= \left(1 - \frac{2}{5}\bar{h}\right)\lambda^2 - \left(\frac{4}{5} + \frac{4}{5}\bar{h}\right)\lambda - \frac{1}{5} = 0$$

$$\text{即 } \lambda^2 - \frac{\frac{4}{5} + \frac{4}{5}\bar{h}}{1 - \frac{2}{5}\bar{h}}\lambda - \frac{1}{5(1 - \frac{2}{5}\bar{h})} = 0$$

要使 $|\lambda| < 1$ 的必要条件为 $|b| < 1 - c < 2$

$$\text{即 } \left| \frac{\frac{4}{5} + \frac{4}{5}\bar{h}}{1 - \frac{2}{5}\bar{h}} \right| < 1 - \frac{1}{5(1 - \frac{2}{5}\bar{h})} = 1 - \frac{1}{5-2\bar{h}} < 2$$

$$\text{而 } 1 - \frac{1}{5-2\bar{h}} < 2 \text{ 成立}$$

$$\frac{5-2\bar{h}-1}{5-2\bar{h}} = \frac{4-2\bar{h}}{5-2\bar{h}}$$

$$\left| \frac{4+4\bar{h}}{5-2\bar{h}} \right| < \frac{4-2\bar{h}}{5-2\bar{h}}$$

$$\text{即 } 2\bar{h}-4 < 4+4\bar{h} < 4-2\bar{h}$$

$$\text{解得 } (-4, 0)$$

五. (1) $u_0 = \int_{-1}^1 |x| dx = 1$

$$u_1 = \int_{-1}^1 |x| x dx = \int_{-1}^0 -x^2 dx + \int_0^1 x^2 dx = 0$$

$$\text{即 } \int_{-1}^1 |x| f(x) dx = \frac{1}{2} [f(-\frac{1}{\sqrt{2}}) + f(\frac{1}{\sqrt{2}})]$$

$$u_2 = \int_{-1}^1 |x| x^2 dx = 2 \int_0^1 x^3 dx = \frac{1}{2}$$

$$u_4 = \int_{-1}^1 |x| x^4 dx = \frac{2}{5} \neq \frac{1}{2} [f(-\frac{1}{\sqrt{2}}) + f(\frac{1}{\sqrt{2}})]$$

$$u_3 = \int_{-1}^1 |x| x^3 dx = 0$$

$$\phi_3(x) = \begin{vmatrix} u_0 & u_1 & 1 \\ u_1 & u_2 & x \\ u_2 & u_3 & x^2 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 1 \\ 0 & \frac{1}{2} & x \\ \frac{1}{2} & 0 & x^2 \end{vmatrix} = \frac{1}{2} x^2 - \frac{1}{4}$$

(2) 由(1)可知 $\phi_3(x) = 0$ 得 $x_1 = -\frac{1}{\sqrt{2}}, x_2 = \frac{1}{\sqrt{2}}$, 故代数精度为3

$$\begin{cases} A_0 + A_1 = 1 \\ -\frac{1}{\sqrt{2}} A_0 + \frac{1}{\sqrt{2}} A_1 = 0 \end{cases}$$

$$A_1 - A_0 = 0 \quad \begin{cases} A_0 = \frac{1}{2} \\ A_1 = \frac{1}{2} \end{cases}$$

六 $A = \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix}$

$$A^T = \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \quad \frac{d}{dt} e^{tA} = \begin{pmatrix} e^t & & \\ (1+t)e^t & e^t & \\ (t^2+2t)e^t & (1+t)e^t & e^t \end{pmatrix}$$

3个 Jordan 块

$$e^{tA} = (e^{tA^T})^T = \begin{pmatrix} e^t & t e^t & t^2 e^t \\ & e^t & t e^t \\ & & e^t \end{pmatrix} = \begin{pmatrix} e^t & 0 & 0 \\ t e^t & e^t & 0 \\ t^2 e^t & t e^t & e^t \end{pmatrix}$$